

Blockchain-ARP Endorsement Benchmark Results

Run: 2026-09-02 17:10-18:46 UTC. N real Fabric orgs (1 peer per org), N = 3, 5, 7, 9. 500 trials each.

Design

- One fixed gateway peer (peer0.org1) generates and signs both the DHCP and switch observations itself, then submits RegisterBinding .
- Endorsement policy: AND of all N orgs (unanimous) - every peer must endorse, not just a majority quorum.
- Metric: wall-clock time of SubmitTransaction itself - propose, broadcast to all other peers, all verify signatures and endorse, order, commit.

Results (commit_latency_ms)

Nodes	Trials	Mean	Median	P95	P99
3	500	2012.31	2012.27	2013.53	2014.30
5	500	2012.82	2012.83	2013.97	2014.58
7	500	2014.81	2014.75	2016.05	2018.67
9	500	2015.75	2015.70	2017.12	2020.02

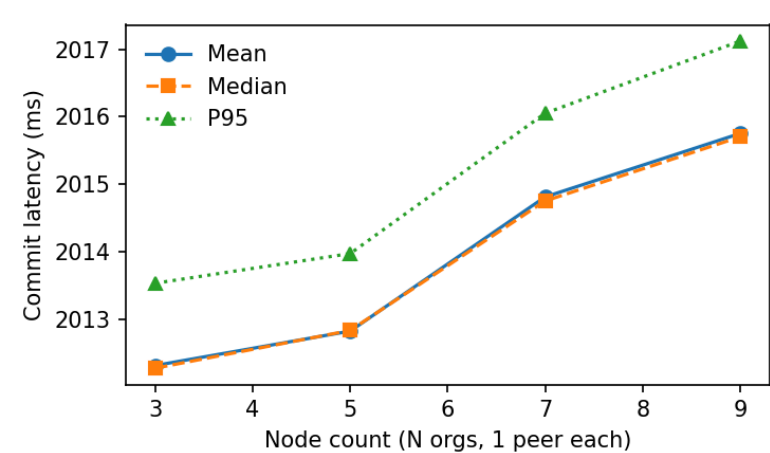
0 failures across all 2000 trials. Latency is flat and clean, small consistent increase per additional peer pair, dominated by Fabric's ~2s block-cutting interval.

Full distribution stats (commit_latency_ms)

Nodes	Min	P10	P25	P50	P75	P90	P95	P99	Max	Mean	Stdev	Range
3	2010.19	2011.32	2011.81	2012.27	2012.77	2013.25	2013.53	2014.30	2019.29	2012.31	0.84	9.11
5	2010.71	2011.87	2012.26	2012.83	2013.26	2013.71	2013.97	2014.58	2019.89	2012.82	0.85	9.19
7	2012.71	2013.78	2014.21	2014.75	2015.30	2015.80	2016.05	2018.67	2022.29	2014.81	1.02	9.58
9	2013.10	2014.65	2015.10	2015.70	2016.29	2016.81	2017.12	2020.02	2023.42	2015.75	1.02	10.32

Standard deviation stays sub-millisecond through N=5 then roughly doubles at N=7/9 (0.84-0.85ms to ~1.02ms); the full min-max range widens similarly (9.11ms to 10.32ms). Both point to slightly more jitter in the endorsement fan-out as peer count grows, consistent with contacting more distinct peer processes per transaction, though the effect is small relative to the ~2s base commit time.

Figure



Files

benchmark/endorsement-results-20260902/